Lab 2 — Exploring the Catalog, Schema and Table Namespace

Description

A guided walkthrough of Catalog Explorer, the UI for browsing data governed by Unity Catalog. You will work down the three-level namespace — catalog, schema, table — inspect the detail available for a table, and run the equivalent SQL commands. This lab is read-only; nothing is created or changed.

Learning Objectives

- Navigate the catalog, schema and table hierarchy in Catalog Explorer.
- Locate column definitions, table properties, history and permissions for a table.
- Distinguish a managed table from an external table.
- Run the SQL equivalents of the Catalog Explorer navigation commands.

Step by Step Guide

1. Open Catalog Explorer. Click Catalog in the sidebar. You do not need running compute to browse objects governed by Unity Catalog.
2. Review the catalogs. The list shows every catalog you can see — the top level of the Unity Catalog metastore. You see an object only if you hold at least the BROWSE privilege on it, so your list may differ from a colleague's.
3. Open a catalog. Click a catalog name to see the schemas it contains. Open the Details tab and note the catalog owner and any comment.
4. Open a schema. Click a schema name to see the tables and views it contains. If the schema also holds volumes, functions or models, click Tables to expand the list.
5. Open a table. Click a table name, then select the Columns tab. Review the column names, data types and any comments.
6. Review the table details. Select the Details tab. Note the table type (managed or external), the storage location and the table properties.
7. Preview the data. Select the Sample Data tab. This tab requires an active compute resource — unlike the metadata tabs, which do not.
8. Review the remaining tabs. Look at Permissions to see who holds which privileges, History for the Delta table's change history, and Insights for the most frequent queries and users over the past 30 days.
9. Filter the list. Return to the schema and use the filter text box to narrow the object list. Note that this filters only the objects currently visible — use the top bar search to search the whole workspace.
10. Run the SQL equivalents. Open the SQL Editor, attach to a SQL warehouse, and run each command in turn, substituting your own object names:

```
SHOW CATALOGS;
SHOW SCHEMAS IN <catalog>;
SHOW TABLES IN <catalog>.<schema>;
DESCRIBE TABLE <catalog>.<schema>.<table>;
DESCRIBE HISTORY <catalog>.<schema>.<table>;
```

11. Compare the two routes. Note that SQL requires SELECT on the object plus USE CATALOG and USE SCHEMA on its parents, so a table you can browse in the UI is not necessarily one you can query.

Outcome

You can navigate the three-level namespace confidently, find the definition, ownership, history and permissions of any table you have access to, and describe the same objects using SQL. You have also seen that visibility in Catalog Explorer and the right to query the data are governed by different privileges.